

Title: The Basics of Renewable Energy

Renewable energy comes from sources that are naturally replenished, such as sunlight, wind, rain, and geothermal heat. Unlike fossil fuels, renewable sources do not release carbon dioxide when used to generate electricity.

Solar energy captures sunlight through photovoltaic cells to produce electricity. Wind power converts kinetic energy from wind into mechanical power through turbines. Hydropower uses flowing water to drive turbines, while geothermal energy harnesses heat from beneath the Earth's surface.

One of the greatest challenges for renewable energy is storage. Since wind and solar depend on weather conditions, large-scale battery systems and energy grids are required to maintain a stable supply.

By reducing greenhouse gas emissions and promoting sustainability, renewable energy plays a key role in fighting climate change and shaping a cleaner future.